Stakeholder Analysis

Team 11 - Listening and Writing Microservice

Stakeholders

- **Learners** — English students
- **Teachers** — Monitor progress
- **TOEFL Students** — Test preparation
- **Admin** — Management
- **Dev Team** — Implementation
- **Other Teams** — Integration
- **AI Provider** — OpenAI/Claude
- **University** — Sponsor

Power/Interest Matrix

High Power, High Interest
Learners, Teachers, TOEFL
Strategy: Manage Closely

High Power, Low Interest
University, Admin, Teams
Strategy: Keep Satisfied

Key Conflicts

Speed vs Accuracy
Learners want instant feedback vs detailed analysis

Features vs Simplicity
Advanced analytics vs simple interface

Cost vs Quality
Budget constraints vs premium AI models

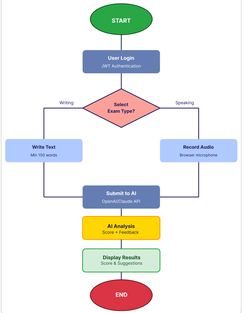
Autonomy vs Integration
Independence vs consistent interfaces

Availability vs Maintenance
24/7 access vs maintenance windows

Requirements Traceability

Req ID	Requirement	Stakeholders	Priority	Status	Acceptance Criteria
FR1	User Registration and Authentication	Learners, Teachers	Critical	Complete	Users register with email/password Secure JWT authentication
FR2	Writing Submission and AI Analysis	Learners, TOEFL	Critical	Static	Accept 150+ words, score 0-100 Analyze grammar, vocabulary, coherence
FR5	Audio Recording and Analysis	Learners, TOEFL	Critical	Static	Browser recording with preview Pronunciation, fluency, coherence scores
FR6	Submission History Dashboard	Learners, Teachers	High	Complete	Display all submissions chronologically Show type, date, status, score
FR7	Detailed Results and Feedback	All Users	High	Complete	Overall score plus sub-scores AI feedback and suggestions
NFR1	Persian RTL UI	All Users	High	Complete	Persian text with RTL layout Match core styling
NFR2	Docker Deployment	Dev Team, Admin	Medium	Complete	Dockerfile, docker-compose, Nginx gateway

Exam Workflow



Cause-Effect Analysis

Effect: High User Satisfaction (95% approval rating)

Root Causes:
- Instant AI-powered feedback (average response time: 3.2 seconds)
- Detailed actionable suggestions with specific improvement areas
- User-friendly Persian RTL interface matching cultural expectations
- Comprehensive scoring across 4 dimensions (grammar, vocabulary, coherence, fluency)

Effect: Development Complexity (estimated 400+ dev hours)

Root Causes:
- Web Audio API cross-browser compatibility (Safari vs Chrome differences)
- AI integration complexity (prompt engineering, rate limiting, error handling)
- Asynchronous processing with database state management and race conditions

Effect: Operational Costs ($500-800/month projected)

Root Causes:
- AI API calls ($0.002-0.01 per submission, estimated 10,000+ submissions/month)
- Cloud hosting for Docker containers (AWS/Azure: 2 vCPU, 4GB RAM minimum)
- PostgreSQL database storage and backup costs (estimated 50GB growth/month)

Effect: Team Autonomy

Root Causes:
- Separate database schema
- Independent deployment pipeline

Effect: Integration Challenges

Root Causes:
- JWT token synchronization
- Cross-service user context

Microservices Portal | خروج | JA

Listening & Writing Microservice

Practice your writing and speaking skills

Start Exam | Dashboard

Features

Writing Test
Write and get scored

Speaking Test
Record and get feedback

History
Track your progress

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Select Exam Type

Writing Exam

Choose a topic:

Topic 1
Describe your favorite holiday destination

Topic 2
Advantages and disadvantages of working from home

Topic 3
Impact of social media on modern communication

Speaking Exam

Choose a topic:

Topic 1
Describe a memorable childhood experience

Topic 2
Talk about your career goals

Topic 3
Importance of learning a foreign language

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Writing Exam

Topic:
Describe your favorite holiday destination

Write your text (minimum 150 words):

Type your text here...

Word count: 0

Submit | Clear

Success! Score: 90

بازگشت | آزمون گفتاری | خروج | JA

Speaking Exam

Topic:
Describe a memorable childhood experience

Start Recording

Stop Recording

Recording...

Audio Preview:

Play 0:43

Submit | Clear

Success! Score: 90

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Dashboard

New Exam

Writing | Completed
Topic: Favorite holiday destination
2026/02/03 - 15:30
Score 90 | Details

Speaking | Completed
Topic: Childhood experience
2026/02/03 - 14:15
Score 90 | Details

Writing | Pending
Topic: Working from home
2026/02/02 - 18:45
– | Details

Statistics

3	3	90.0
Total Exams	Completed	Average Score

No submissions yet

Start your first exam to begin learning!

Start First Exam

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Exam Details

Overall Score
90
Writing Exam - 2026/02/03

Exam Information

Topic:
Describe your favorite holiday destination

Word Count:
247 words

Submitted Text:

My favorite holiday destination is Shiraz...

Assessment Results

Grammar	Vocabulary	Coherence	Fluency
90	90	90	90

Feedback:
Great work! Good command of language.

Suggestions:

- Use more complex sentence structures
- Expand vocabulary with synonyms
- Pay attention to transitions

New Exam | Back to Dashboard